Explore 1.1

When does a transformation of the roots reproduce the same cubic?

Roots of polynomial equations: linear-fractional transformations of roots

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Colour key: green = verification check | blue = alternative method.

Explore 1.1

The polynomial $x^3+x-3=0$ has roots α,β,γ. If $\dfrac{a\alpha+1}{\alpha-b}$, $\dfrac{a\beta+1}{\beta-b}$, $\dfrac{a\gamma+1}{\gamma-b}$ are the roots of another cubic, find the conditions on a and b that make this second cubic *identical* to the first.

Setup. For $x^3+0\,x^2+x-3=0$ the symmetric functions are

$$\alpha+\beta+\gamma=0, \qquad \alpha\beta+\beta\gamma+\gamma\alpha=1, \qquad \alpha\beta\gamma=3.$$

Every root is sent to its image under the single map

$$T(x)=\frac{ax+1}{x-b}.$$

The new cubic is the same as the old one precisely when the *set* of images $\{T(\alpha),T(\beta),T(\gamma)\}$ is again $\{\alpha,\beta,\gamma\}$ — that is, when T **permutes the three roots**.

Step 1 — the cubic satisfied by the images. Put $y=T(x)$ and invert to express x in terms of y:

$$y(x-b)=ax+1 \implies x(y-a)=by+1 \implies x=\frac{by+1}{y-a}.$$

Each image y therefore makes $x=\dfrac{by+1}{y-a}$ a root of $x^3+x-3=0$. Substituting and clearing the denominator $(y-a)^3$ gives the cubic in y:

$$(by+1)^3+(by+1)(y-a)^2-3(y-a)^3=0.$$

Expanding and collecting powers of y:

$$\underbrace{(b^3+b-3)}_{\text{coeff } y^3}y^3+\underbrace{(3b^2-2ab+9a+1)}_{\text{coeff } y^2}y^2+\underbrace{(a^2b-9a^2-2a+3b)}_{\text{coeff } y^1}y+\underbrace{(3a^3+a^2+1)}_{\text{coeff } y^0}=0.$$

Note that the leading coefficient is exactly $f(b)$, where $f(x)=x^3+x-3$.

Step 2 — match to the original. For this to be the *same* cubic as $y^3+0\,y^2+y-3=0$, the four coefficients must be in the ratio $1:0:1:-3$. Dividing by the leading coefficient gives three conditions:

$$\text{(E1)}\;\; 3b^2-2ab+9a+1=0, \qquad \text{(E2)}\;\; \frac{a^2b-9a^2-2a+3b}{b^3+b-3}=1, \qquad \text{(E3)}\;\; \frac{3a^3+a^2+1}{b^3+b-3}=-3.$$

A cleaner route to (E1) is the partial-fraction split $a\alpha_i+1=a(\alpha_i-b)+(ab+1)$, so that

$$T(\alpha_i)=a+\frac{ab+1}{\alpha_i-b}, \qquad \text{hence} \qquad \sum_i T(\alpha_i)=3a+(ab+1)\sum_i\frac{1}{\alpha_i-b}=3a-(ab+1)\frac{f'(b)}{f(b)}.$$

Setting $\sum_i T(\alpha_i)=0$ and using $f'(b)=3b^2+1$ reproduces (E1) after clearing $f(b)$. The same split shows the whole image set is controlled by the single number $ab+1$ — a fact that becomes decisive in a moment.

Step 3 — solve the system. Eliminating a between (E1)–(E3) collapses the system completely: the resultant in b has b^3+b-3 as its only common factor, and the reduced relations are

$$\boxed{b^3+b-3=0} \qquad \text{and} \qquad \boxed{a=-\frac{b^2+1}{3}}.$$

The first condition says b is itself a root of the original cubic. Using $b^3+b=3$, i.e. $b(b^2+1)=3$, so $b^2+1=\frac{3}{b}$, the second condition simplifies to

$$a=-\frac{1}{3}\cdot\frac{3}{b}=-\frac{1}{b}, \qquad \text{i.e.} \qquad ab=-1.$$

So the only values that force the coefficients to agree are: **b a root of $x^3+x-3=0$, together with $a=-1/b$.**

Step 4 — interpretation: every such solution is degenerate. These two conditions are exactly the two ways the transformation breaks down.

(i) $ab=-1$ destroys the map. The matrix of T is $M=\begin{pmatrix} a & 1 \\ 1 & -b \end{pmatrix}$ with $\det M=-(ab+1)$, which vanishes when $ab=-1$. Concretely, with $a=-1/b$,

$$T(x)=\frac{-x/b+1}{x-b}=\frac{-(x-b)/b}{x-b}=-\frac{1}{b} \qquad \text{(a constant)}.$$

(ii) b a root sends one image to infinity, since the denominator α_i-b then vanishes for that root.

So the three "images" are really $\left\{-\frac{1}{b},\,-\frac{1}{b},\,\infty\right\}$: two of them coincide and one runs off to infinity. They do not form a genuine cubic at all. Hence

there is no non-degenerate choice of a,b for which the two cubics genuinely coincide.

Alternative — why it must fail (Möbius symmetry argument). For T to genuinely permute the roots it must be a non-identity symmetry of the three-element root set, realised by a *real* linear-fractional map. But x^3+x-3 has discriminant $-4(1)^3-27(3)^2=-247<0$, so it has *one real root and a complex-conjugate pair.* A real T sends reals to reals, so it must fix the lone real root. The only non-trivial options are then:

- a 3-cycle, which needs $\operatorname{tr}^2 M=\det M$, i.e. $(a-b)^2=-(ab+1)$, rearranging to $a^2-ab+b^2+1=0$. Since $a^2-ab+b^2=\left(a-\frac{b}{2}\right)^2+\frac{3}{4}b^2\geq 0$, the left side is always ≥ 1 — *impossible for real a,b*;
- an involution swapping the conjugate pair, which needs $\operatorname{tr} M=0$, i.e. $a=b$ — but $a=b$ is incompatible with the system above (it would force $b^2=-1$).

With both symmetry types ruled out over $\mathbb{R}$, the algebra has nowhere to land except the degenerate locus of Step 4. (A cubic with *three real* roots could admit a genuine real 3-cycle, so the obstruction here is entirely the complex-conjugate pair.)

Check. Symbolic elimination of the matching system returns the Gröbner basis $\{\,3a+b^2+1,\ b^3+b-3\,\}$ exactly — so $b^3+b-3=0$ and $a=-(b^2+1)/3=-1/b$ is the *complete* solution set, with no branch omitted. Taking the real root $b\approx 1.21341$ gives $a=-1/b\approx -0.82412$ (and indeed $-(b^2+1)/3\approx -0.82412$ ✓). The three images then evaluate to -0.82412, -0.82412 and an undefined value ($\to\infty$): they collapse, confirming degeneracy. The determinant $-(ab+1)=0$ ✓.

Conclusion. Three symmetric-function conditions on only two parameters leave the system over-determined, and its sole solutions sit on the degenerate locus $b^3+b-3=0,\ a=-1/b$, where T collapses to a constant and one image escapes to infinity. So *no finite, non-degenerate pair (a,b) makes the two cubics the same* — the lone real root and the complex-conjugate pair leave no room for a genuine root-permuting symmetry of the required form.